ACS APPLIED POLYMER MATERIALS

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Enhancing Thin Film Morphology and Conductivity Through Serial Doping of Conjugated Thiophene-Based Copolymers

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Cite This: *ACS Appl. Polym. Mater.* 2025, 7, 12083−12092 Read Online

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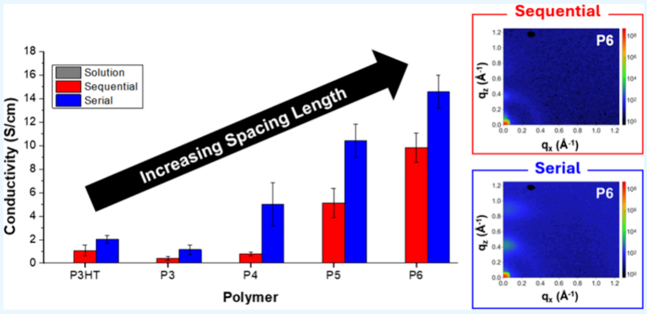
ABSTRACT: Chemical doping has been widely used to enhance the electrical performance of conjugated polymer thin films for organic electronic devices with specific attention given to dopant molecule selection, modifications of copolymer design to enhance dopant interactions, and the doping process itself. Here, a set of expanded core thiophene-based copolymers are doped via solution, sequential, or "serial" doping processes. Serial doping, which utilizes solution and sequential doping methods in succession, was utilized to investigate how the use of a two-step doping method impacts the doping efficiency, thin film conductivity, and morphology. Results show that serial doping with 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ) increases the conductivity of all of the thiophene-based copolymers relative to either one-step solution or sequential doping processes. This increase in conductivity of serial-doped films is not attributed to greater levels of incorporation of F4TCNQ; rather, the serial-doped thin films display more ordered crystalline microstructures, despite the pristine films having amorphous character. This indicates that serial doping enhances the conductivity by inducing crystallization. This study also shows that as the spacing length between alkyl side chains of the expanded core copolymers is increased, the conductivity and the ordering of crystallites increases. Overall, this work provides insight into how copolymer design and the doping method impact conductivity through complex and intertwined relationships between dopant integration, effective electron transfer, and thin film microstructure.



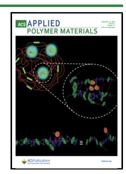
KEYWORDS: *conjugated polymers, conductivity, chemical doping, morphology, thiophene, thin films*

■ INTRODUCTION

Conjugated polymers are widely studied for their use in thermoelectric devices, sensors, and printed electronics due to their flexible nature, low cost, processability, and use of naturally abundant elements.[1−3] The electrical performance of conjugated polymers can be optimized or tailored by using molecular dopants, where energetic offsets result in electron transfer that ultimately produces charge carriers. The polymer/dopant pair that has been studied most extensively is poly(3-hexylthiophene) (P3HT) and 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ), which feature well-aligned highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) levels, respectively, which allows for effective p-type doping.[4−10] However, it is widely appreciated that the method used to incorporate the dopant significantly impacts processability, morphology, and conductivity.[7,8,10−15]

The most facile method of doping is solution doping, which relies on the direct blending of separate solutions of the polymer and the dopant. This method is proven to be effective in enhancing thin film conductivity; however, large amounts of dopant are required, and charge transfer leads to a significant amount of chain aggregation in solution, which can be detrimental to film morphology.[8,10] Therefore, a sequential doping process is often used to prevent aggregation. This method requires that a pristine polymer film is created first, and then the dopant is added using a marginal solvent to swell the polymer chains, allowing the dopant to diffuse into the film.[8,10,13,16] The use of the pristine polymer film allows the morphology/chain alignment to be set prior to dopant integration and thus avoids the aggregation that arises from charge transfer during solution doping. Sequential doping is considered superior to solution doping because it preserves film morphology and requires less dopant while maintaining or enhancing the conductivity. Within this approach, it has also been shown recently that binary solvent mixtures can be used to strategically tailor polymer−solvent interaction energy in order to optimize film conductivity.[16]

Received: July 25, 2025
Revised: August 14, 2025
Accepted: August 15, 2025
Published: August 29, 2025



https://doi.org/10.1021/acsapm.5c02754
ACS Appl. Polym. Mater. 2025, 7, 12083−12092

A few doping strategies that modify the sequential doping process have been developed to further improve the electrical performance. For instance, Fontana et al. demonstrated that evaporation-based sequential doping of P3HT with F4TCNQ yields electrical conductivities and degrees of doping that are similar to those realized by solution-based sequential doping.[7] They note that while the evaporation-based doping is more advantageous for large-area films, solution-based sequential doping is preferable for films with micrometer scale thicknesses.[7] In another use of evaporation-based sequential doping, Patel et al. showed that vapor deposition of (tridecafluoro-1,1,2,2-tetrahydrooctyl)trichlorosilane (FTS) into thin films of poly(2,5-bis(3-tetradecylthiophen-2-yl)-thieno[3,2-*b*]thiophene) (PBTTT-C_{14}) increased the thermoelectric performance (enhanced Seebeck coefficient and power factor) compared to the films that were sequential-doped via solution immersion.[17] Similarly, Vijayakumar et al. showed that higher thermoelectric power factors and high electrical conductivity could be achieved by a sequence of sequential doping steps using solutions of progressively higher dopant concentrations.[18] They showed that this so-called incremental concentration doping (ICD) preserved the microstructural ordering present in the initial films.[18]

Other studies have also investigated the impact of using a method that utilizes successive solution and sequential doping steps. For example, Chai et al. used a method called "Sequential-Twice-Doping" in which P3HT was solution-doped with F4TCNQ (at 7 mol %) and the resulting film was then sequentially doped through immersion of the solution-doped film in an iron chloride ($FeCl_3$) solution.[11] The films created by this method exhibited an enhanced conductivity compared to films made by either solution- or sequential-doping only. Their assessments of morphology showed that the solution doping step doped the amorphous regions while the sequential doping step improved doping efficiency while maintaining the size of crystalline domains.[11] Similarly, Yoon et al. used what they termed "hybrid doping" in which thiophene-based polymers, including P3HT, were doped in solution with F4TCNQ at various levels (5−20 wt %), cast into a thin film, and then sequentially doped by immersion in an F4TCNQ solution.[14] This hybrid doping method resulted in higher film conductivity compared to solution or sequential doping alone, which also maximized the thermoelectric power factor.[14] These doping methods that combine the use of solution and sequential doping show significant promise in terms of enhancing electrical performance but thus far have been limited to studies that use dopant concentrations that establish modest or high levels of solution doping, where aggregation of polymer chains due to charge transfer is known to be prevalent.

More recently, it has been reported that doping efficiency may be improved by altering the repeat design of conjugated polymers in a way that increases the "space" between solubilizing side chains in order to enhance polymer-dopant interactions.[16,19−21] An example of this strategy of altering sterics along the backbone comes from Li et al., who showed that the incorporation of fused-ring thienothiophene spacer units into poly(quaterthiophene) enhanced doping efficiency by reducing the steric demand along the backbone, which ultimately led to higher conductivity in sequential-doped thin films.[19] Kim et al. showed that statistical copolymers of 3-hexylthiophene and thiophene exhibited conductivities 100 times higher than that of P3HT due to the formation of polymer-dopant-polymer cocrystals with short packing distances.[21] This cocrystallization is facilitated by the reduced side chain density as it enables the dopant to more closely interact with the polymer backbone.[21]

Herein, we report "serial" doping, a two-step doping process consisting of both solution and sequential doping, of a series of thiophene-based copolymers that are systematically altered to reduce side chain density. (We choose the term serial doping to clarify the process as having two doping steps that are performed in sequence, rather than hybrid doping, which can signify a blend or mixture.) A distinct aspect of this study is the use of low concentrations of F4TCNQ dopant in the solution doping step (1 mol %, which equates to <0.4 wt %, depending on the copolymer used) to avoid aggregation. We investigate the impact that copolymer design and doping method have on thin film conductivity, dopant integration, and morphology. We find that serial doping is a superior doping method compared to either solution or sequential doping methods. Specifically, the results show that for this series of thiophene-based copolymers, serial doping produces conductivities that are higher than either of the parent doping processes and promotes the formation of highly ordered crystallites.

■ EXPERIMENTAL SECTION

Materials. 2,5-bis(trimethylstannyl)thiophene (97%, Sigma), 2,5-bis(trimethylstannyl)thieno[3,2-*b*]thiophene (97%, Sigma), 5,5′-dibromo-2,2′-bithiophene (98%, TCI), 5,5′′-dibromo-2,2′:5′,2′′-terthiophene (97%, Sigma), 2-bromo-3-hexylthiophene (97%, Sigma), 3-hexylthiophene-2-boronic acid pinacol ester (95%, Sigma), 3-hexylthiophene (99%, Sigma), tetrakis(triphenylphosphine)palladium(0) ($Pd(PPh_3)_4$, 99%, Sigma), palladium(II) acetate ($Pd(OAc)_2$, 98%, Sigma), potassium carbonate (99.7%, Fisher), neodecanoic acid (NDA, Sigma), magnesium sulfate ($MgSO_4$, 99.5%, Alfa Aesar), 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ, 99%, Ossila), toluene (99.85%, extra dry, Thermo Scientific), diethyl ether (anhydrous, Fisher), ethyl acetate (99%, Fisher), hexanes (98.5%, Fisher), dimethylformamide (DMF, 99.8%, Thermo Scientific), N,N-dimethylacetamide (DMAc, 99.5%, extra dry, Thermo Scientific), methanol (99%, Fisher), acetone (99%, Fisher), chloroform (98%, Fisher), chlorobenzene (99.8%, anhydrous, Sigma), and graphite conductive adhesive (aqueous based, Alfa Aesar) were used as received. N-bromosuccinimide (NBS, 99%, Sigma) was recrystallized in water and vacuum-dried prior to use.

Synthetic Procedures. 2,5-dibromo-3-hexylthiophene and the series of expanded core comonomers were synthesized using previously reported procedures.[16] P3HT and the expanded core copolymers were synthesized via direct arylation polymerization (DArP) as described in brief in the Supporting Information, and full details can be found in our recent work.[16]

Polymer Characterization. The ^1H NMR spectrum of each copolymer, acquired using a Varian VNMRS 500 MHz NMR, is presented in Figures S1−S5. Gel permeation chromatography (GPC) was used to determine the number-average molecular weight (M_n) and dispersity (Đ) of each copolymer based on universal analysis. An Agilent 1260 Infinity II system was used for these measurements with a mobile phase of THF at 25 °C and a flow rate of 1 mL/min. Solutions for GPC measurements were made at a concentration of 4 mg/mL in THF by first heating at 50 °C for 15 min (or until fully dissolved) before cooling to room temperature. Each solution was filtered (without back-pressure) through a 0.2 PTFE filter prior to injection. The resultant molar masses and dispersities are summarized in Table S1.

DSC Measurements. Polymer samples were dried overnight in a vacuum oven. DSC samples with sample weights of ∼3−7 mg were made in aluminum hermetic pans, and an empty aluminum hermetic pan was used as a reference. A TA Instruments Q-2000 DSC instrument was used to measure the thermograms from −50 to 300

https://doi.org/10.1021/acsapm.5c02754
ACS Appl. Polym. Mater. 2025, 7, 12083−12092

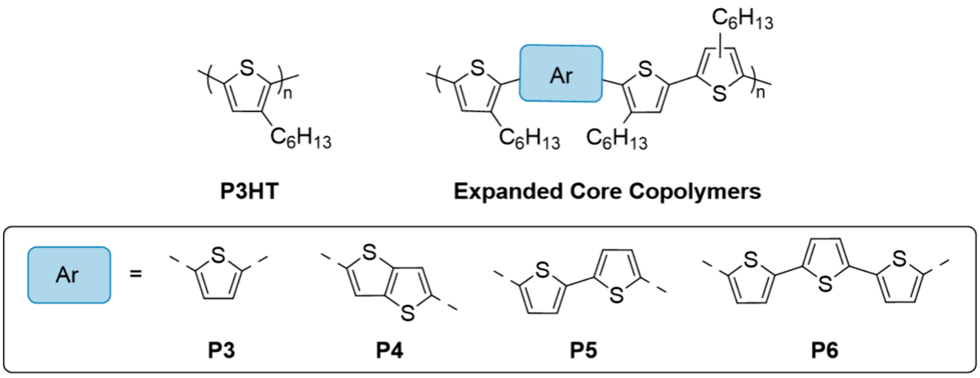


Figure 1. Structures of **P3HT** and the expanded core copolymers containing spacing units of thiophene (**P3**), thieno[3:2,b]thiophene (**P4**), 2,2′-bithiophene (**P5**), and 2,2′:5′,2′′-terthiophene (**P6**).

°C at a rate of 10 °C/min. Both heating and cooling cycles were measured twice. The first cycle was used to erase any thermal history of the polymers, and analyses were performed on the second heating and cooling cycle.

Thin Film Preparations. *Solution Doping.* Solutions of 1 mol % F4TCNQ (relative to the total number of moles of thiophene repeat units in the chain, which equates to 0.30−0.40 wt % depending on the copolymer used) were made by mixing separately prepared solutions of the copolymer and F4TCNQ in chlorobenzene. For each solution, the final copolymer concentration after mixing was 30 mg/mL. The resultant polymer/dopant solution was then heated to 70 °C prior to deposition. Glass slides (1 in × 1 in) were cleaned with KimWipes, washed with acetone and methanol, and dried with a stream of dry N_2 gas. The polymer/dopant solution (200 μL) was spin-cast dynamically onto the clean glass slide at 1000 rpm for 40 s. The spin-cast polymer film was allowed to dry for at least 10 min before adding contact points for conductivity measurements (see Conductivity Measurements). After adding contact points at the corners, the films were dried for an additional hour prior to any measurement.

Sequential Doping. The copolymer was dissolved in chlorobenzene (30 mg/mL) at 70 °C while F4TCNQ was dissolved in acetone at 1 mg/mL at room temperature. Clean glass slides were prepared as described above. Then, the polymer was spin-cast onto the clean glass slide by depositing 200 μL of polymer solution dynamically at 1000 rpm for 40 s. Immediately after spin-casting, the thin film was submerged in the dopant solution for 30 s. The resulting sequentially doped copolymer films were allowed to dry for at least 10 min before adding contact points, and then dried for an additional hour, as described above.

Serial Doping. The same procedure described for solution doping was used, and immediately after that, a second sequential doping step followed. The sequential doping procedure was performed by submerging the solution-doped thin film in the dopant solution for 30 s. For each step, dopant concentrations were set to be equivalent to those of the single-step doping processes: In other words, solution doping used an F4TCNQ solution of 1 mol % F4TCNQ and the sequential doping step used an acetone solution at 1 mg/mL F4TCNQ. The resulting serial-doped copolymer films were allowed to dry for at least 10 min before adding contact points, and then dried for an additional hour, as described above.

Conductivity Measurements. Conductive graphite adhesive was used to add contact points at each corner of the thin films. A Keithley 2450 Sourcemeter was used to perform current−voltage (I−V) sweeps from 0 to 1 V. Voltage was applied at the contact points across one edge and the current was measured across the contact points on the opposite edge in accordance with the Van der Pauw method, which requires measurement along each edge of the film.[22−24] A set of 3 replicate films were measured for each doping method. The sheet resistance (R_s) was calculated using the slope of the I−V curve and the Van der Pauw equation.[24] R_s and the film thicknesses, t, obtained from AFM scratch test measurements (described in Supporting Information), were used to calculate the film resistivity, ρ. Conductivity, σ, was subsequently calculated as $\sigma = \rho^{-1}$.

Thin Film UV−vis-NIR Measurements. UV−vis-NIR spectra over the wavelength range 350 < λ (nm) < 2000 were acquired from thin films using a Cary 5000 UV−vis-NIR Spectrophotometer. This was done for as-cast and all doped thin films. A background spectrum acquired from a clean glass slide was used to baseline correct the spectra acquired for thin films and the corrected spectra were then normalized by film thickness. The absorbance ratio of F4TCNQ$^-$:polymer was calculated using the absorbance at λ = 868 nm, which corresponds to the anionic form of F4TCNQ, F4TCNQ$^-$, and the absorbance at the λ_{max} for each of the respective polymers. Solution-based UV−vis measurements of the expanded core copolymers are shown in our prior work.[16]

GIWAXS Measurements. Thin films were prepared as described above; however, an ozone treated silicon wafer substrate (critical angle = 0.22°) was used in place of the glass. Measurements were performed by using a Xenocs Xeuss 3.0 instrument. A Xenocs GeniX3D Cu source with a wavelength of 1.54 Å was used at an incident angle of 0.2°. The detector distance was set at 55 mm, and each measurement was performed for 5 min. The resultant 2D images were used to produce 1D spectra based on a line cut in the out-of-plane and in-plane directions, as indicated.

RESULTS AND DISCUSSION

Copolymer Synthesis and Characterization. A series of thiophene-based copolymers with expanded cores were used to elucidate the impact that repeat unit design has on morphology and conductivity, with an emphasis on the doping method. Inspired in part by the work of Li et al.,[19] the expanded cores were designed with varying thiophene-based "spacing units" to increase the distance between hexyl side chains in the comonomers, which allows for more effective dopant integration within the copolymer due to reduced steric demand.[16,19,20] The expanded core copolymers feature spacing units of thiophene (**P3**), thieno[3:2,*b*]thiophene (**P4**), 2,2′-bithiophene (**P5**), and 2,2′:5′,2′′-terthiophene (**P6**), as shown in Figure 1. **P3HT** with a regioregularity of 93% was made via AB polymerization and is used as a benchmark polymer. The copolymers **P3**-**P6** have a regioregularity of 50% due to the AA +BB polymerization of the symmetric expanded core comonomers with the asymmetric comonomer 2,5-dibromo-3-hexylthiophene. The molar masses of the (co)polymers range from 6 to 16 kg/mol (Table S1). As described in our recent work, based on these molar masses, each polymer is well-beyond its effective conjugation length;[16] therefore, any subsequent changes seen in optoelectronic measurements are believed to be attributed to repeat unit design and/or polymer self-assembly.

https://doi.org/10.1021/acsapm.5c02754
ACS Appl. Polym. Mater. 2025, 7, 12083−12092

To first investigate the impact that spacing units have on thermal properties, DSC measurements were performed to identify the glass transition temperature (T_g), crystallization temperature (T_c), and melting temperature (T_m) of these polymers. The full DSC thermograms are depicted in Figures S7−S11, and the values of T_g, T_c, and T_m are summarized in Table 1. First, **P3HT** does not exhibit a distinguishable T_g despite reports of P3HT having a weak T_g in the range of ∼5−22 °C.[25,26] However, clear T_c and T_m are observed, indicating that this regioregular polymer is semicrystalline, as expected. The value of T_m = 196 °C is consistent with reported values of T_m for P3HT, which range from ∼190 to 230 °C depending on regioregularity and molar mass.[27−30] The thermograms acquired for the expanded core copolymers show T_g transitions; however, these copolymers lack crystallization and melting transitions. This indicates that copolymers **P3**-**P6** are amorphous in nature, which may be due to their inherent regio-irregularity. All have a higher T_g compared to the values reported for P3HT, with **P4** having the highest T_g due to the backbone rigidity resulting from its fused ring. The T_g values for the other expanded core copolymers follow the trend of **P3** < **P5** < **P6**, which is consistent with the expected impact of increasing the spacing unit length (decreasing the side chain number density), which allows for more π-stacking interactions between the polymer chains.

Table 1. T_g, T_c, and T_m for P3HT and the Expanded Core Copolymers

Polymer[a]	T_g (°C)	T_c (°C)	T_m (°C)
P3HT	n.m.	161	197
P3	29	-	-
P4	55	-	-
P5	35	-	-
P6	42	-	-

[a]All values obtained from DSC thermograms acquired at 10 °C/min n.m. - not measured.

Preparation and Characterization of Doped Copolymer Thin Films. Various doping methods have been developed to enhance the conductivity of conjugated polymer thin films, with the most prevalent methods being solution doping and sequential doping. These two doping methods are used here along with serial doping to assess the impacts on conductivity and morphology. As shown in Figure 2, solution doping entails combining premixed solutions of polymer and dopant and then spin-casting this polymer/dopant solution onto the substrate to obtain solution-doped thin films. On the other hand, sequential doping involves taking a pristine polymer thin film and then, in a separate step, integrating the dopant by soaking the polymer film in a dopant solution. This method requires the use of a marginal solvent (with respect to the polymer) to ensure effective dopant integration without the dissolution of the film. In this study, acetone is utilized because our prior work shows that it has a moderate polymer−solvent interaction energy with **P3HT** and the expanded core copolymers, which optimizes the conductivity realized by sequential doping.[16] While solution and sequential doping methods each add the dopant in a single step, serial doping subjects a solution-doped thin film to a second doping

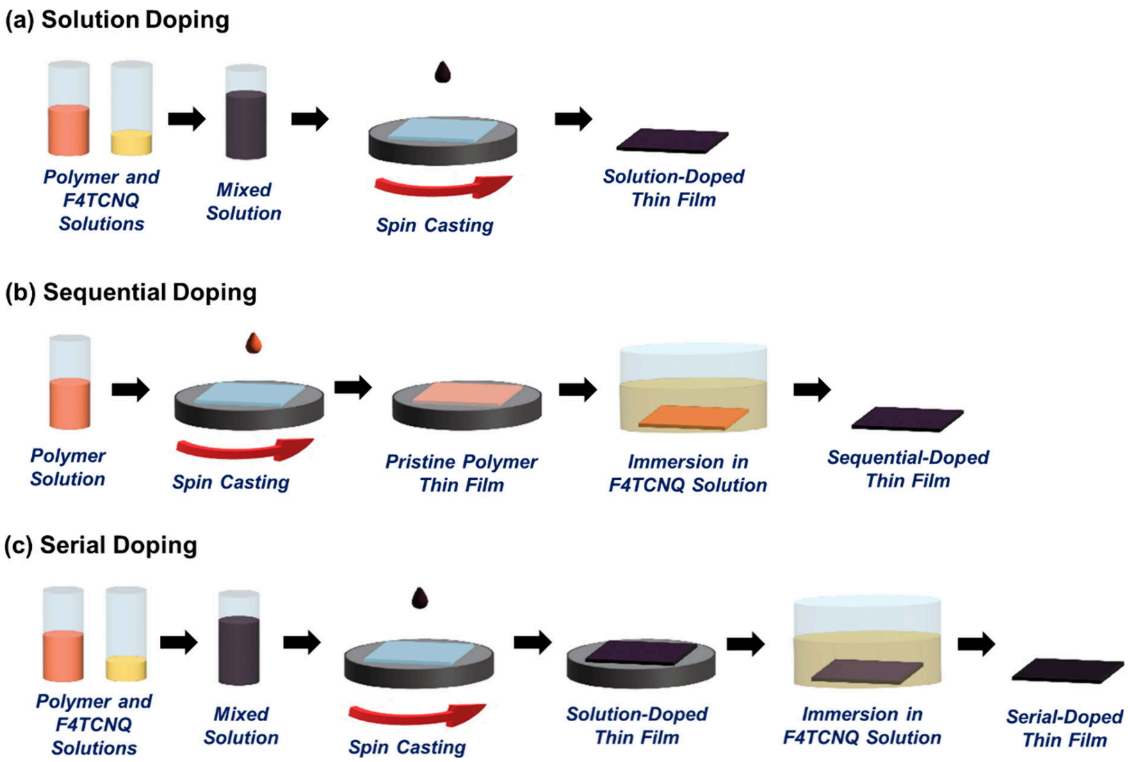


Figure 2. Schematic drawings illustrate (a) solution, (b) sequential, and (c) serial doping methods. Solution-doped thin films are cast from a solution made by mixing separate solutions of 30 mg/mL polymer and 1 mol % F4TCNQ in chlorobenzene at 70 °C. For sequential doping, a 30 mg/mL polymer solution in chlorobenzene at 70 °C is used to cast a pristine polymer film, and then the film is submerged in a 1 mg/mL F4TCNQ solution in acetone for 30 s. Serial doping involves first creating the solution-doped thin film, followed by a second doping step in which the solution-doped thin film is submerged in a 1 mg/mL F4TCNQ solution in acetone for 30 s.

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ACS Appl. Polym. Mater. 2025, 7, 12083−12092

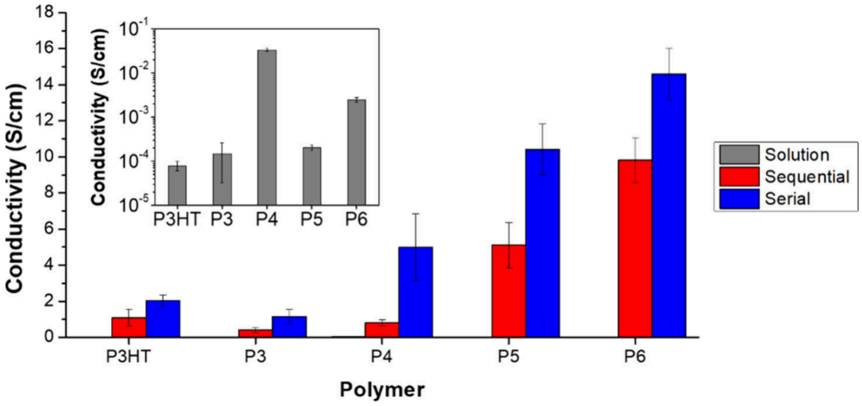

Figure 3. Conductivity of solution-, sequential-, and serial-doped thin films of **P3HT** and each of the expanded core copolymers. The inset shows the solution-doped thin film measurements on a logarithmic scale to visualize the data effectively. For solution doping, films were cast from a polymer solution having 1 mol % F4TCNQ. With sequential doping, the polymer thin films were submerged in a 1 mg/mL F4TCNQ solution in acetone for 30 s. Serial doping used a two-step process in which a solution-doped film was made and then submerged in a 1 mg/mL of F4TCNQ solution in acetone for 30 s. Clearly, the increase in conductivity resulting from serial doping exceeds what would be expected from a purely additive result.

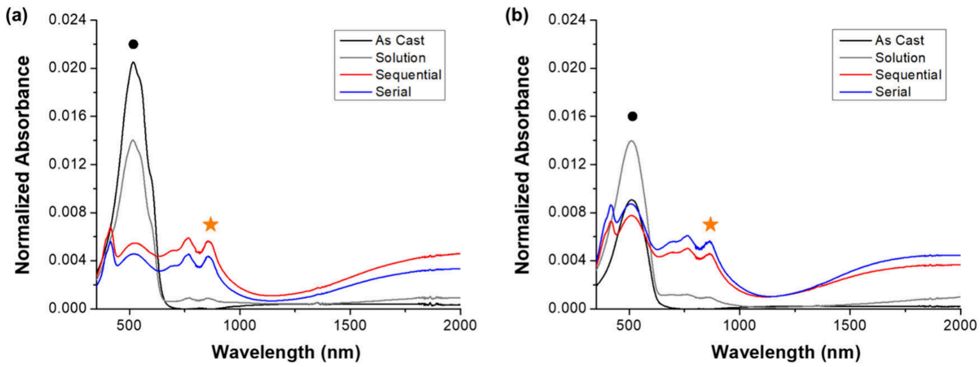

Figure 4. UV−vis-NIR spectra of (a) **P3HT** and (b) **P3** as-cast films and films doped via solution, sequential, and serial doping methods. The spectra are normalized by film thickness, which makes the absorbance signal proportional to concentration. The black circles indicate the λ_{max} of **P3HT** and **P3** at 517 and 512 nm, respectively, while the orange star indicates the peak attributed to $F4TCNQ^{-}$ (868 nm) that is used for analysis. The broad absorbance appearing at $\lambda > 1200$ nm signifies polaron formation.

step that is analogous to sequential doping by soaking that film in a dopant solution (that was prepared using a marginal solvent). These processes are depicted in Figure 2.

Each of the copolymers, along with **P3HT**, were doped using these three methods, and thin film conductivities were subsequently measured. It should be appreciated that even though consistent solution concentrations and spin-casting processes were used, the thin films have inherently different thicknesses, ranging from 62 to 300 nm (depending on the copolymer and doping method, Table S2), because the polymers have different solubilities (manifests as different Hansen Solubility Parameters[16]). First and as shown in the inset of Figure 3, the solution-doped thin films have low conductivities that are in the range of $\sim 10^{-4}-10^{-1}$ S/cm. (The exact values are summarized in Table S3.) This is attributed to the small amount of dopant used (1 mol % with respect to the number of thiophene units present in solution). There is a clear trend that as the size of spacing unit increases from thiophene to bithiophene to terthiophene, the conductivity also increases (**P3** ≤ **P5** < **P6**). This is attributed to the larger spacing units allowing for more effective dopant integration due to the decreased number density of side chains along the backbone. Notably, solution-doped **P4** films have a significantly higher conductivity compared to the other copolymers. This is ascribed to both the fused-ring structure promoting π-interactions and the reduced steric demand offered by the repeat unit design, which was also observed by Li et al. in their study of sequentially doped thin films of poly(quaterthiophene) having thienothiophene spacing units (PQT-TT).[19] A more encompassing description of the thin film microstructure and its relation to conductivity is presented later. As seen in Figure 3, the conductivity measured for all polymer films prepared by sequential doping is three orders-of-magnitude higher than their solution-doped counterparts. This increase in conductivity is largely due to a significant increase in the amount of dopant integrated into the thin film upon sequential doping compared to that resulting from solution doping. This is attributed to the low level of dopant used with solution doping (1 mol % F4TCNQ) and confirmed by measurements of the amount of dopant in the film, which are discussed in more detail later. The conductivities measured from sequentially doped films again show that as the spacing unit size is increased, there is an increase in conductivity: **P3** < **P5** < **P6**; however, unlike the solution-doped films, the sequentially doped **P4** film no longer exhibits the highest conductivity. This arises due to changes in film microstructure, as evidenced by GIWAXS measurements, which are also presented later.

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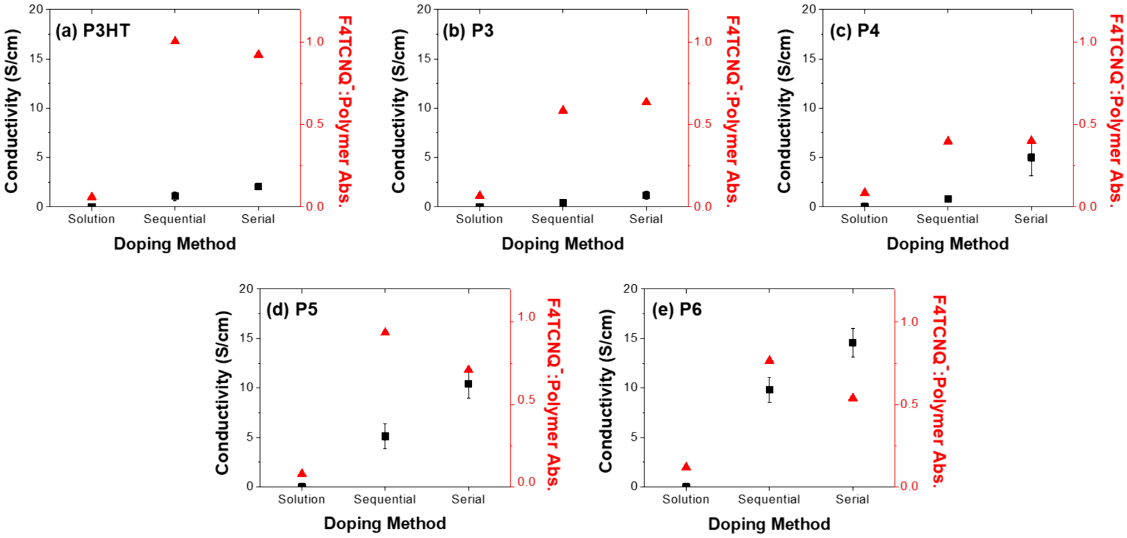

Figure 5. Comparison of conductivity and F4TCNQ$^-$:Polymer absorbance ratio for thin films created by each doping method: (a) **P3HT**, (b) **P3**, (c) **P4**, (d) **P5**, and (e) **P6**. The ratio of F4TCNQ$^-$:Polymer was calculated using the absorbance at 868 nm for F4TCNQ$^-$ and the absorbance at the λ_{max} for each of the respective polymers. Common axis scales are used to enable comparisons.

Figure 3 also clearly shows that serial doping results in a higher conductivity for each of the copolymer and **P3HT** films compared to that with sequential doping. The solution-doped thin films start with a very low conductivity, but the subsequent step of sequential doping results in the conductivity surpassing that of sequentially doped films. This indicates that the two steps of the serial doping process are not additive, which suggests that this increase in conductivity is not directly caused by the amount of dopant introduced into the thin film. Rather, we contend that the increase in conductivity is largely impacted by the microstructure of the film, and this idea is discussed in more detail later.

One indication of effective doping is the presence of anionic F4TCNQ (F4TCNQ$^-$) in doped systems, whether in solution or in thin film. The formation of F4TCNQ$^-$ signifies an electron transfer from the conjugated polymer backbone to F4TCNQ; therefore, higher levels of F4TCNQ$^-$ indicate that more doping events have occurred. UV−vis-NIR measurements were used to evaluate the amount of dopant, specifically F4TCNQ$^-$, present in the thin films. Figure 4 shows spectra from as-cast, solution-, sequential-, and serial-doped thin films of **P3HT** and **P3** as examples. These spectra are normalized by film thickness, which makes the normalized absorbance directly proportional to the concentration. The neutral polymer peaks (labeled with black circles) of **P3HT** and **P3** are observed at 517 and 512 nm, respectively. The presence of F4TCNQ$^-$ is observed in the range of 728−955 nm, with the local maximum at 868 nm identified in each spectrum by the orange star.[5,6,16] Notably, the solution-doped **P3HT** and **P3** thin films have the lowest absorbance of F4TCNQ$^-$, indicating that less doping occurs, and as expected, these films have the lowest conductivities. The low concentrations of F4TCNQ$^-$ associated with the solution-doped thin films are also observed for the other expanded core copolymers, and those spectra are presented in Figures S12−S14. Compared to solution-doped thin films, the absorbance of the neutral polymer peaks for sequential- and serial-doped films decreases as the F4TCNQ$^-$ peak increases, indicating that these doping methods result in a greater occurrence of electron transfer. Although Figure 4 shows that sequentially doped **P3HT** films have a higher F4TCNQ$^-$ absorbance than the corresponding serial-doped films, the conductivity of sequentially doped **P3HT** films is lower than that of the serial-doped **P3HT** films. On the other hand, the serial-doped **P3** thin films have the highest concentration of F4TCNQ$^-$, which is consistent with those films having the higher conductivity. However, this small increase in F4TCNQ$^-$ for serial-doped **P3** films does not account for the ∼3× increase in conductivity measured for serial-doped **P3** thin films compared to sequential-doped films, which was seen in Figure 3. As shown in Figures S12−S14, **P4** and **P5** films show a higher absorbance of F4TCNQ$^-$ with serial doping compared to that with sequential doping, while **P6** films show a higher absorbance of F4TCNQ$^-$ with sequential doping. The inconsistency in the amount of F4TCNQ$^-$ present in serial-doped films further indicates that the amount of F4TCNQ$^-$ present in the thin film is not the sole contributor to conductivity, and it may also suggest that not every charge carrier pair that is formed contributes to conduction, as demonstrated by Pingel and Neher in their studies of P3HT and F4TCNQ.[31]

Additional insight into the doping effectiveness was generated from these spectra by calculating the ratio of absorbance of F4TCNQ$^-$ to that of neutral polymer. The absorbance of F4TCNQ$^-$ at 868 nm (identified by the orange star in Figure 4) and the absorbance of each polymer at their respective λ_{max}, which was determined from their respective as-cast spectra, were used in this calculation. Figure 5 shows the correlation between conductivity and the ratio of the F4TCNQ$^-$:neutral polymer for solution-, sequential-, and serial-doped thin films. For solution-doped thin films of **P3HT** and each expanded core copolymer, the F4TCNQ$^-$:Polymer is very low, which is consistent with the low conductivities (∼10^{-4}−10^{-1} S/cm) that were seen from the inset in Figure 3. Compared to solution-doped films, films made using sequential doping or serial doping have higher F4TCNQ$^-$:Polymer absorbance ratios and higher conductivities; however, there is no clear trend between the absorbance ratio and conductivity for sequential- and serial-doped films. For **P3HT**, **P3**, and **P4** the ratio of F4TCNQ$^-$:Polymer remains relatively constant, despite these films exhibiting an increase in

https://doi.org/10.1021/acsapm.5c02754
ACS Appl. Polym. Mater. 2025, 7, 12083−12092

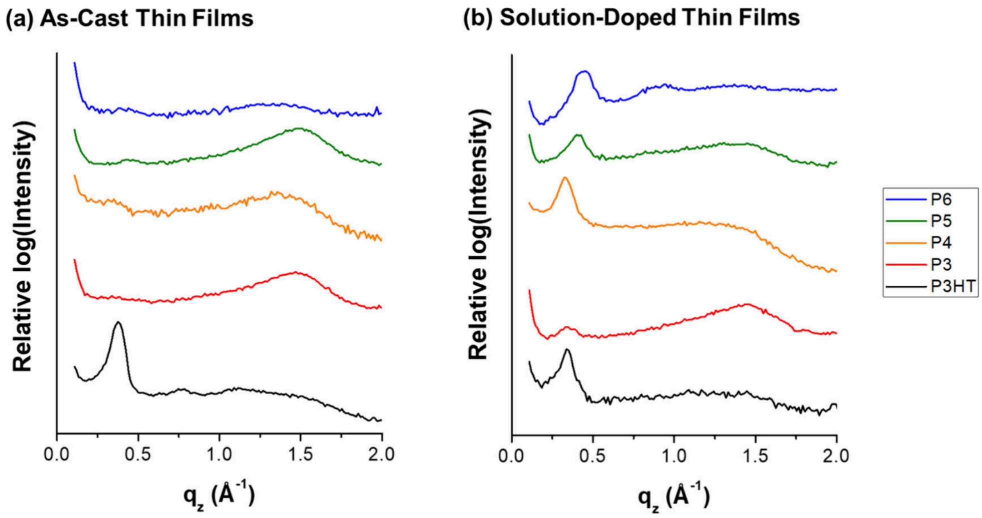

Figure 6. Out-of-plane 1D GIWAXS profiles of (a) as-cast and (b) solution-doped thin films. The *y*-axis is offset on a logarithmic scale to facilitate a comparison. Corresponding 2D images are shown in Figure S15 and S16.

conductivity from sequential doping to serial doping. With **P5** and **P6**, there is a decrease in the $F4TCNQ^-$:Polymer ratio when comparing sequentially doped to serially doped films, even though an increase in conductivity is observed. These findings indicate that the difference in conductivity between sequential and serial doping is not solely driven by the amount of $F4TCNQ^-$ generated during doping and suggests that not all of the $F4TCNQ^-$ present in the thin film contributes to charge transport.

The thin films were characterized using GIWAXS to gain insight into how the doping method and copolymer design impact ordering. Figure 6 shows the 1D GIWAXS profiles in the out-of-plane direction for the as-cast and solution-doped thin films of each polymer. We focus on the edge-on orientation because in-plane conductivity measurements were performed, and it has been shown that edge-on crystallites facilitate higher in-plane conductivity due to intermolecular charge transport occurring along the π-stacking direction.[32,33] Analogous in-plane GIWAXS profiles are shown in Figures S19−S23 and the corresponding 2D images are presented in Figure S15 and Figure S16, respectively.

For the as-cast films, **P3HT** has a well-defined (100) diffraction peak along with a weak (200) diffraction peak in the out-of-plane direction. The narrowness of the (100) peak along the azimuthal direction in the corresponding 2D image indicates that it adopts a predominantly edge-on orientation.[33,34] This finding is consistent with DSC measurements that indicate that **P3HT** is semicrystalline. The as-cast thin films of the expanded core copolymers (Figure 6a) appear to be essentially amorphous (consistent with DSC measurements exhibiting only a T_g), as there are no crystalline peaks with **P3** and only very weak (100) peaks with **P4**, **P5**, and **P6**. In addition, an amorphous halo is observed between 1 and 2 $Å^{-1}$ for each of the expanded core copolymers and **P3HT**. For solution-doped **P3HT** thin films, the crystalline peak is less defined, indicating that the films are more disordered than the as-cast films, which is likely due to aggregation of chains that occurs upon electron transfer. Furthermore, and as summarized in Table 2, the out-of-plane *d*-spacing calculated from the (100) diffraction peak is larger for solution-doped **P3HT** thin films (18.6 Å) compared to that for as-cast **P3HT** films (16.7 Å). This indicates that the dopant is integrated within the lamellae, thus increasing the lamellar distance.[12,35] With the expanded core polymers, (100) crystalline peaks become more apparent in the solution-doped thin films, suggesting that the integration of the F4TCNQ dopant induces crystallization. This phenomenon has been reported for P3HT doped with low concentrations of F4TCNQ.[12,36] Notably, solution-doped films of **P4** have a more well-defined crystalline peak than the other copolymers that feature expanded cores and exhibit the highest conductivity among the films made using this doping method. This indicates that presence of edge-on crystallites results in higher conductivity compared to the other solution-doped copolymer thin films.

Table 2. Out-of-Plane *d*-Spacing of As-Cast and Doped Thin Films

Polymer	Out-of-plane *d*-spacing (Å)[a]			
	As-cast[b]	Solution	Sequential	Serial
P3HT	16.7	18.6	19.3	20.1
P3	-	18.6	20.1	19.3
P4	-	19.3	21.0	18.6
P5	-	15.6	17.9	18.6
P6	-	13.9	17.9	14.7

[a]Calculated from the corresponding 1D GIWAXS profiles using the (100) diffraction peak. [b]As-cast films of P3−P6 are amorphous; thus, there are no *d*-spacing values to report.

Figure 7 shows the out-of-plane 1D profiles and 2D GIWAXS images of sequential- and serial-doped thin films, with the 2D image axes scaled to highlight diffraction peaks associated with the edge-on orientation (full scale images are shown in Figure S17 and S18). Sequential doping is known to maintain the morphology of the pristine polymer thin films after doping,[8,10,37] and this phenomenon is seen here with sequentially doped **P3HT**. Specifically, sequentially doped P3HT has a clear (100) peak with weak higher-order reflections, similar to those of the pristine films. With sequentially doped **P3**, **P4**, and **P5**, the 1D profiles, which are presented in Figure 7a, are largely unchanged compared to their as-cast counterparts (Figure 6a) and the 2D images show Debye−Scherrer rings demonstrating that crystallites adopt all possible orientations with respect to the plane of the film. A weak (100) diffraction peak emerges from the 1D profile of **P6**

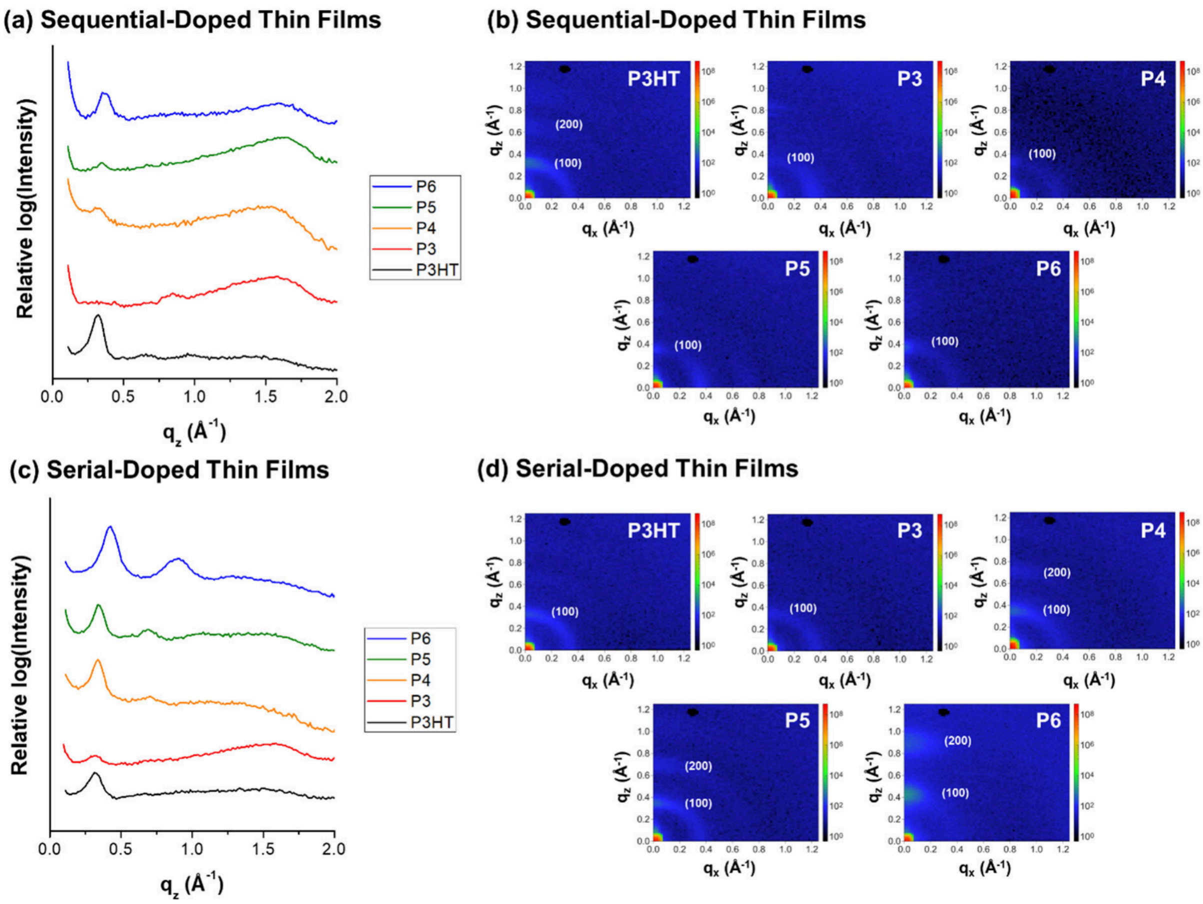

Figure 7. GIWAXS measurements are represented as (a) out-of-plane 1D profiles and (b) 2D images from sequential-doped films along with analogous (c) 1D profiles and (d) 2D images acquired from serial-doped thin films. The 2D images are scaled to emphasize the diffraction peaks associated with edge-on orientation. Full scale 2D images are shown in Figure S17 and S18. Serial doping improves the microstructural ordering for **P3HT** and each copolymer (**P3**-**P6**) compared with sequential doping.

suggesting that the incorporation of the dopant promotes some ordering within the film. Also, each of the sequentially doped thin films shows an increase in the out-of-plane *d*-spacing compared to that of solution-doped films (Table 2), suggesting that more dopant is integrated into the film. This contention is consistent with both the large increase in the F4TCNQ$^-$:Polymer absorbance ratio and the increase in conductivity observed for sequentially doped thin films compared to films prepared by solution doping.

With serial doping (Figures 7c and 7d), GIWAXS measurements suggest that **P3HT** exhibits a microstructure similar to that of its sequentially doped films. **P3** now shows the presence of a weak (100) peak, which indicates a slight increase in order due to the second doping step; however, the diffuse (100) peak seen in the 2D image indicates that there is no preferred orientation with respect to the substrate. On the other hand, **P4**, **P5**, and **P6** all show features that support a significant increase in order. Specifically, the out-of-plane 1D profiles obtained from each of these copolymer films display (100) diffraction peaks, similar to the sequentially doped films; however, weaker (200) peaks are also observed. The appearance of a higher-order peak indicates a significant increase in structural ordering due to the second doping step. The 2D images add additional support to the contention of a more ordered microstructure compared to that of their sequentially doped counterparts, as multiple crystalline peaks are seen for **P4**, **P5**, and **P6**. This is most clearly seen with **P6**, which has very intense, narrow features in the out-of-plane direction, indicating a preferred crystalline orientation of edge-on. This improved microstructural order is believed to be the predominant reason for the higher in-plane conductivity observed for serial-doped thin films compared to sequential-doped films, as seen in Figure 3. Additionally, with serial doping, the out-of-plane *d*-spacing decreases with spacing length according to **P3HT** > **P3** > **P4** = **P5** > **P6**. This indicates that the expanded core designs reduce steric demand and allow for more intimate interaction between F4TCNQ and repeat units along the backbone. This contention is supported by the work of Kim et al., who showed that statistical copolymers of 3-hexylthiophene and thiophene exhibited shorter packing distances than P3HT due to the reduced side chain density in the copolymers, which allowed for cocrystallization of the polymer and dopant.[21]

■ CONCLUSIONS

A series of thiophene-based copolymers featuring expanded cores were used to assess the impact that spacing length between alkyl side chains has on thin film morphology and conductivity, with particular emphasis on the effect of the method of doping. Serial doping, a two-step process consisting of both solution doping and sequential doping steps, increased thin film conductivity in comparison to sequential doping even though the dopant concentration used in the first step was low. This increase in conductivity with serial doping was not

https://doi.org/10.1021/acsapm.5c02754
ACS Appl. Polym. Mater. 2025, 7, 12083−12092

attributed to the amount of dopant or F4TCNQ$^-$ present in the film. Rather, this increase in conductivity is largely impacted by enhancements in crystalline content caused by dopant integration. This is supported by the emergence of distinct (100) diffraction peaks in the serially doped copolymer thin films, despite the pristine thin films being largely amorphous in character. Moreover, higher-order peaks were also present in the films subjected to serial doping that were made from copolymers with larger spacing units (**P4**, **P5**, and **P6**). This finding suggests that the use of serial doping can overcome the predominantly amorphous character of a conjugated polymer thin film, leading to a more ordered microstructure, which, in turn, enhances conductivity. With serially doped **P3HT** films, the enhancement of ordering and conductivity appears to be minimal, likely due to its intrinsic crystallinity.

In addition, this study shows that as the spacing length between alkyl side chains in the expanded core copolymers is increased, the conductivity and the extent of microstructural ordering increases. Overall, this study provides key insights into how conductivity is affected by the complex interplay among dopant integration, efficient electron transfer, and thin film microstructure in thiophene-based polymer thin films. Ultimately, the serial doping process leads to more ordered microstructures despite the pristine polymer thin films exhibiting largely amorphous character, which indicates that polymers that are intrinsically semicrystalline may not be needed to achieve higher conductivities.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsapm.5c02754.

> Synthetic procedures associated with P3HT and core expanded copolymers; ^{1}H NMR characterization; GPC traces and molar mass data; AFM thickness measurements; Conductivity of doped thin films, DSC thermograms; UV−vis-NIR spectra of doped thin films; 2D GIWAXS images of as-cast and solution-doped thin films; In-plane 1D GIWAXS profiles (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

ANL acknowledges support from the National Defense Science and Engineering Graduate (NDSEG) Fellowship program. SMK also acknowledges support from the National Science Foundation (Award #2204396) and the U.S. Department of Energy's Office of Energy Efficiency and Renewable Energy (EERE) under the award number DE-EE0009177 provided to the University of Tennessee − Oak Ridge Innovation Institute (UT-ORII). The Scintillation Materials Research Center at the University of Tennessee is thanked for access to the Cary 5000 Spectrophotometer. GIWAXS measurements were enabled by the Major Research Instrumentation program of the National Science Foundation under Award No. DMR-1827474.

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